

THARUKI WIJAYARATHNE

Teaching Assistant - Department of Civil Engineering, University of Peradeniya

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EDUCATION

- Bachelor of Science of Engineering Honours (B. Sc. Engineering)** 2021 - 2025
Specialized in Civil Engineering
Faculty of Engineering, University of Peradeniya
◦ **GPA: 4.00/4.00**, First Class Honors [\[transcript\]](#)
◦ **Top of the Class in Civil Engineering | Rank (1/143)**
◦ **Top of the Engineering Batch 2019 | Rank (1/483)**
◦ Complete degree program was conducted and assessed in **English Medium** and accredited by the **Washington Accord**.
Sri Lanka
- G. C. E Advanced Level Examination (University Entrance Examination)** Aug 2019
3 Distinction (A) passes for Combined Mathematics, Physics, and Chemistry
Top 0.7% in Kandy District, Top 0.4% in Sri Lanka out of 19000+ candidates
Z-Score: 2.6672
Kandy, Sri Lanka
- MOOCs**
 - **CSI ETABS Masterclass** by Asian Institute of Engineering (16 CPD hours)(Completed Apr 2025).
 - **Engineering Drawing and 3D Modelling using AutoCAD** by the Institute of Computer Aided Engineering (Completed Mar 2021).
 - Diploma in Information Technology (International) by ESOF Metro Campus (Pearson Assured) (Completed Feb 2020).

PUBLICATIONS

- [J.1] **R.M.T.D Wijayarathne, K.M.S.M Jayarathna, K.K Wijesundara, Validation of a Simplified Performance-Based Wind Design Framework for Tall Buildings Using CAARC Wind Tunnel Data**, Elsevier Structures Journal, Manuscript submitted for publication, 2026 (Under Review)
- [C.1] **R.M.T.D Wijayarathne, K.M.S.M Jayarathna, K.K Wijesundara, Towards a Performance-Based Wind Design Framework for High-Rise Buildings Using Time-History Analysis**, Annual Sessions 2026, Society of Structural Engineers, Sri Lanka-[\[link\]](#)
- [C.2] **R.M.T.D Wijayarathne, K.M.S.M Jayarathna, K.K Wijesundara, Time-History-Based Evaluation of Wind-Induced Serviceability Response in High-Rise Buildings**, Proceedings of Peradeniya University International Research Symposium and Exposition (iPURSE) 2026-[\[link\]](#)

SELECTED RESEARCH PROJECTS

- Performance-Based Design of High-Rise Buildings under Wind Loading (Final Year Research Project)**
Tools: ETABS, OpenSees, Python | Nov 2024–Aug 2025 | [Project Report](#) |
 - Developed and validated a high-rise wind response model using ETABS (LTHA) and STKO/OpenSees (NLTHA).
 - Generated and applied wind time histories/buffeting loads and benchmarked response against CAARC wind tunnel results.
 - Compared dynamic response metrics (top displacement/acceleration) and assessed modeling/damping assumptions for serviceability-focused design.
 - Supervised by Prof. K.K. Wijesundara
- Serviceability Wind Assessment of a 63-Storey High-Rise (Colombo, NCD Consultants (Pvt) Ltd)**
Tools: ETABS, Python | Nov 2025–Jan 2026 | [Project Report](#) |
 - Performed serviceability wind checks for a 63-storey tower designed by NCD Consultants (Pvt) Ltd.
 - Evaluated wind-induced structural response against comfort/serviceability criteria (e.g., peak/RMS acceleration and drift-related limits as applicable).
 - Summarized critical response levels and supported design decision-making for wind serviceability performance.
- Yaka Bridge Pier Retrofitting - Finite Element Model Development**
Tools: MIDAS FEA NX, SolidWorks | Dec 2025–Feb 2026 | [Project Report](#) |

- Built a detailed 3D solid FE model of the existing pier and retrofit scheme, including geometric transitions and key components for assessment.
- Supported retrofitting design verification through stiffness/response evaluation and model-based structural assessment outputs.
- Prepared analysis-ready geometry/model organization for downstream loading and checking workflows.
- **2D and 3D Finite Element Analysis of Existing Sewer Line Under Proposed Seven-Storey Raft Foundation**
Tools: MIDAS FEA NX | April 2026–June 2026 | [Project Report](#) |
 - Conducted 2D plane strain and full 3D soil-structure interaction analysis to assess the impact of the proposed seven-storey residential building on the existing 1200 mm diameter sewer line at Kandy Railway Station. Developed detailed model incorporating irregular raft, DCM piles, layered soil, sand cushion, and staged construction simulation.
 - Evaluated pipe ovaling, hoop stresses, principal stresses, contact pressure, and additional settlement induced by building loads.
 - Supervised by Prof. K.K. Wijesundara
- **Response Spectrum Seismic Analysis of a 146.5 m Composite Chimney, Norochcholai**
Tools: MIDAS FEA NX | July 2026–Present | [Project Report](#) |
 - Performing modal and response spectrum analysis in MIDAS FEA NX to evaluate the seismic performance of the 146.5 m composite chimney at the Lakvijaya (Norochcholai) coal power plant under site-specific earthquake loads.
 - Assessing key structural responses (base shear, tip displacements, modal behavior) and providing design recommendations for improved seismic resilience and code compliance.
 - Supervised by Prof. K.K. Wijesundara

HONOURS AND AWARDS

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| • Ceylon Development Engineering Prize
<i>Awarded for the best performance in Civil Engineering</i> [Certificate] | Aug 2025 |
| • Mr. Helarisi Abeyruwan Gold Medal
<i>Awarded for outstanding performance in Civil Engineering</i> [Certificate] | Aug 2025 |
| • A. Thuraiajah Prize for Geotechnics
<i>Faculty of Engineering, University of Peradeniya</i> [Certificate] | Aug 2025 |
| • H.B. De Silva Prize for Surveying in Civil Engineering
<i>Faculty of Engineering, University of Peradeniya</i> [Certificate] | Aug 2025 |
| • Prof. Nimal Seneviratne Award for Best Civil Engineering Project
<i>Structural and Materials Sub-discipline</i> [Certificate] | Aug 2025 |
| • Seismic Performance of Building Design Competition 3.0-Bangkok, Thailand
<i>Team member representing the University of Peradeniya in an international seismic design challenge</i> | Jul 2025 |
| • ASCE Sri Lanka Section Student Chapter National Quiz Competition – 1st Place
<i>Team BuildBrains; National-level Civil Engineering Quiz Competition</i> [Certificate] | Jun 2024 |
| • Best Performing Team – Evolucionar'22 Orientation Programme
<i>Team Leader; presented on the structural assessment of ancient structures in Anuradhapura</i> | Jun 2023 |

EXPERIENCE

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| • Teaching Assistant
Department of Civil Engineering, University of Peradeniya | Nov 2025 - Present
Peradeniya, Sri Lanka |
| ◦ Conduct tutorial and instructional sessions in Structural Design, Environmental Engineering Design, Structural Analysis, and Computer-Aided Design (CAD) for undergraduate civil engineering students. | |
| ◦ Support students in applying design codes, analytical methods, and software tools to solve practical engineering problems. | |
| • Trainee Civil Engineer
Stems Consultants (Private) Ltd | Jul 2024 - Sep 2024
Colombo, Sri Lanka |
| ◦ Assisted in preparing structural drawings and design calculations under senior engineer supervision. | |
| • Trainee Civil Engineer
Senura Civil Engineering (Pvt) Ltd | July 2023 - Oct 2023
Bingiriya, Sri Lanka |
| ◦ Assisted in day-to-day site supervision and monitored construction activities to ensure compliance with drawings and specifications. | |

SKILLS

- **Core Engineering:** Structural Analysis and Design, Hydrological Modelling, Site Planning, Roadway Design, Project Management
- **Structural & Design Software:** ETABS, SAP2000, Tekla Tedds, Tekla Structures, OpenSees, MIDAS FEA NX
- **CAD & Civil Design:** AutoCAD, Civil 3D
- **Hydrology & Hydraulics:** HEC-RAS, HEC-HMS
- **GIS & Spatial Analysis:** ArcGIS
- **Programming Languages:** MATLAB, Python, C#

EXTRA CURRICULAR

- **ENGEX 2025 – Engineering Exhibition (University of Peradeniya):** Constructed and exhibited a scaled physical model of *Jethawanaramaya*; also presented a numerical modeling study of high-rise building behavior, explaining structural concepts and analysis results to academic and industry audiences.
- **Community Volunteering (Rotaract, Arunella, and Voice of Youth):** Engaged in community- and education-focused initiatives, including teaching and mentoring school children through outreach programs, rural education support, and youth development activities.
- **Post-Cyclone Damage Assessment (Central Province, Sri Lanka):** Participated in a university-led post-cyclone field assessment program, contributing to on-site inspection of school buildings, documentation of observed damage, and preliminary structural safety and hazard identification.

REFERENCES

1. **Prof. K.K. Wijesundara , Ph.D. (University of Pavia, Italy), M.Sc. (University of Pavia, Italy), B.Sc.Eng. (Peradeniya, Sri Lanka), AMIESL**
Professor & Head, Department of Civil Engineering
University of Peradeniya, Peradeniya, Sri Lanka.
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2. **Dr. Sahan Bandara , Ph.D. (Swinburne University of Technology, Australia), B.Sc.Eng. (Peradeniya, Sri Lanka), AMIESL**
Senior Lecturer, Department of Civil Engineering
University of Peradeniya, Peradeniya, Sri Lanka.
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3. **Dr. S. P. C. Perera , Ph.D. (Texas Tech University, USA), M.Sc. (Texas Tech University, USA), B.Sc.Eng. (Peradeniya, Sri Lanka)**
Senior Lecturer, Department of Engineering Mathematics
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